An ultra-specialist



Giant pandas occupy a very specialized ecological niche, as their diet consists mainly of bamboo. Bamboo is a poor food source, low in protein and high in cellulose. Pandas can digest only a small proportion of what they eat, which means they have to eat a lot of bamboo—as much as 28 lb (12.5 kg) each day—and forage for up to 14 hours a day. It is unclear why pandas have become so dependent on bamboo, but some zoologists suggest it is because it is an abundant and reliable food source, and pandas are not skilled predators.

Pandas eat different parts of bamboo plants according to the seasons. In late spring, they prefer the first green shoots. They eat leaves at other times of the year, and stems in winter when little else is available. Pandas have evolved muscular jaws and a pseudothumb to manipulate bamboo stems. Their digestive tract is inefficient at processing large quantities of plant material because it remains similar to that of its carnivorous ancestors, although digestion is helped by the bacterial fauna in their gut.

See also: Competitive exclusion principle • Field experiments • Optimal foraging theory • Animal ecology • Niche construction • Halting biodiversity loss